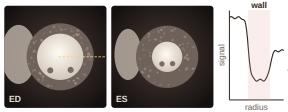
Figure 1 | Two-track evidence map for regional strain from routine cine CMR

ESTIMATION TRACK

a Estimation chain: cine supplies image cues; correspondence is inferred, strain is defined

panel a: observed in cine · inferred by the estimator · defined by convention

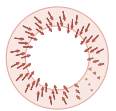
1 · IMAGE FORMATION (I)
Cine bSSFP, mid short axis



Observed: blood–myocardium borders (conspicuous), intramural texture (weak), time. Confounders: coils, through-plane motion, partial volume, artefact.

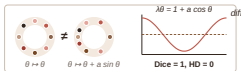
inferred

2–3 · CORRESPONDENCE (E) (R)
Estimated displacement u(X, t)



$$\hat{\varphi}_{0\to t} = \arg\min_{\varphi} \mathcal{D}(I_0, I_t \circ \varphi) + \alpha\,\mathcal{R}(\varphi)$$

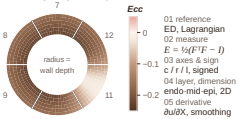
data term does not identify the map; the prior selects
tracking · registration · FE/biomechanics · supervised
· self-supervised · compact basis



Identical masks admit different material maps: contour agreement cannot certify correspondence. Temporal scheme (reference, sequential, groupwise) adds a further assumption.

differentiate

4 · STRAIN DEFINITION
Green–Lagrange Ecc, pixelwise

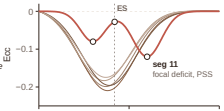


01 reference
ED, Lagrangian
02 measure
$E = \tfrac{1}{2}(F^TF - I)$
03 axes & sign
c / r / l, signed
04 layer, dimension
endo-mid-epi, 2D
05 derivative
∂u/∂X, smoothing

Ring radius is wall depth (endo inner, epi outer), not the base→apex axis of a 17-segment bullseye; septum left. Engineering −0.20 is GL −0.18: a definitional 0.02 gap.

aggregate

5 · REGIONAL REPORT
Segmental Ecc(t), mid ventricle



06 region
AHA mid 7–12

07 temporal, seg 11

○ at ES	−0.03
○ peak systolic	−0.08
○ post-systolic	−0.12

08 aggregation order

min of mean	−0.17
mean of minima	−0.18

Specifications 01–08 together define one regional estimand; ○ readouts and 08 (6 segments) are computed from the curves.

Failure points, each testable on its own: (I) **input sufficiency** — re-image the same motion finer · (E) **estimation** — vary α with images fixed · (R) **representation** — project the known field onto the basis

VALIDATION TRACK

b Validation targets: each reference supports one claim; results as reported

panel b: DENSE · tagging · cine FT · cine DL · comparator · scar

Known motion Morales 2021; Buoso 2023
✓ capacity & technical error
⊗ not alone: in-vivo accuracy

prescribed peak strain (truth) · **DeepStrain error (mean ± SD)**



Dice 0.82 (lowest in thin scar) · displacement 1.0 ± 0.9 mm
Slice averages only: no scar-region error · infarct EF 49%

Paired DENSE / tagging
✓ material-sensitive in-vivo agreement
⊗ not alone: ground truth (reference error)

Militaru 2021 · cine FT vs tagging, 3 vendors



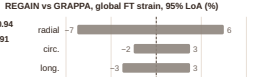
Cao 2018 · DENSE vs tagging
Ecc 0.778

StrainNet (Wang 2023) vs conventional FT; ref DENSE
ES global Ecc 0.72 vs 0.87
ES AHA segment Ecc 0.48 vs 0.75

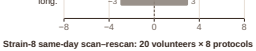
Militaru n = 61 at 3 T; StrainNet 59 cine tests, end-systolic.
Range = spread across vendors; ○ = comparator.

Precision & reconstruction Yoon 2023; Bell 2025
✓ precision at the tested scale
⊗ not alone: accuracy — repeatable can be biased

REGAIN vs GRAPPA, global FT strain, 95% LoA (%)



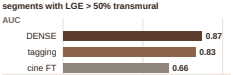
Strain-8 same-day scan–rescan: 20 volunteers × 8 protocols



Mean global differences near zero; radial limits widest.
Arrows mark the reported bound, not a measured CV.

Tissue (LGE) Kihlberg 2020
✓ scar-associated dysfunction
⊗ not alone: pointwise displacement error

segments with LGE > 50% transmural



116 suspected CAD; 34 with qualifying scar.
Ranks scar sensitivity, not material-motion accuracy.

Outcome & readers Bello 2019; Masutani 2023
✓ prognostic / diagnostic association
⊗ not alone: clinical utility

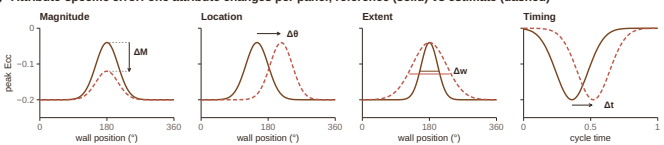
discrimination (chance = 0.5)



C-index (survival) and AUC (classification) share a 0.5 chance level but are not directly comparable.

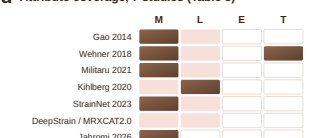
Masutani: 53 patients, 846 segments, 30% peak Err cut-off.
Utility also needs a decision, threshold and cost of error.

c Attribute-specific error: one attribute changes per panel; reference (solid) vs estimate (dashed)

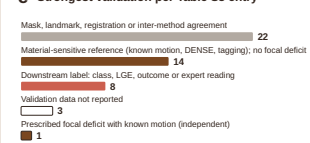


Equal-mean test: a uniform field and a focal field can share one global mean (−0.18); recovering the mean is not a pass.
Report bias and LoA, localisation, extent and timing error, subject-level distributions, correlated segments.

d Attribute coverage, 7 studies (Table 3)



e Strongest validation per Table S3 entry



a, c: conceptual schematics, no study data (a5 readouts computed from the drawn curves). b, d, e: values as reported, transcribed from the manuscript (Tables 2–3, Table S3, §3.6, §4); d coded from Table 3 wording; not pooled — quantities differ.
M magnitude · L location · E extent · T timing. Ecc/Err/Ell, circumferential/radial/longitudinal strain. ES, end-systole; PSS, post-systolic shortening; FT, feature tracking; GL, Green–Lagrange; HD, Hausdorff distance; LoA, limits of agreement; WMA, wall-motion abnormality.